Rahul Kumar

+91-7607089856 | prathamskt@gmail.com | linkedin.com/in/rahul-kumar-445b2b227 | github.com/prarahul

SUMMARY

Data analyst and applied ML engineer (B.Tech CS + M.Tech AI) who builds end-to-end analytics systems: SQL data models, **Power BI dashboards**, ETL pipelines and ML applications. Validated **100,000+ records** in industry and delivered projects spanning **4.6M-record supply chain analytics**, drug-discovery prediction and real-time IoT monitoring.

EXPERIENCE

Famous Innovation — Data Analyst Intern Apr 2026 – Jul 2026

- Validated **100,000+ records** for accuracy, completeness and consistency, surfacing trends and anomalies that fed client-facing business insights.
- Built a geo-location application for real-time field tracking and operational monitoring, combining structured data workflows with location data.
- Ran root-cause analysis on data and application issues and fixed them, improving data quality and operational efficiency.
- Translated stakeholder requirements into data-driven solutions to support project execution and delivery.

Maharashtra Metro Rail Corporation Ltd. — Full Stack Developer Intern Jul 2024 – Aug 2024

- Built a PHP/MySQL Inward-Outward Reception Management System for visitor, parcel and document data, with validation, search, status tracking and reporting.

CODTECH IT Solutions — Data Analyst Intern May 2024 – Jul 2024

- Cleaned and analysed datasets with Python, SQL, Pandas and NumPy to identify trends, relationships and anomalies for decision support.
- Built reusable data-processing workflows that standardised recurring analyses and reduced manual effort.

PROJECTS

SupplyIQ 2.0 — Supply Chain Analytics Platform **[GitHub]** PostgreSQL, SQL, Power BI, Docker

- Designed a **PostgreSQL dimensional model (6 dimension, 5 fact tables)** holding **~4.6M synthetic records** generated with Python, Pandas, SQLAlchemy and Faker; containerised with Docker Compose.
- Wrote SQL analytical views covering revenue, profitability, inventory, procurement, shipment delivery, returns and supplier performance.
- Built an executive **Power BI dashboard** over **1M+ orders and shipments** tracking revenue, profit margin, on-time delivery, inventory and year-over-year trends.

MolecuLens AI — Drug Bioactivity Prediction **[GitHub] [Live Demo]** Python, Scikit-learn, RDKit, Streamlit

- Trained a **Random Forest** model on **416 ChEMBL compounds** to predict **EGFR (L858R/T790M)** inhibitor bioactivity; **scaffold-based 5-seed cross-validation** gave **MAE 0.611 and R^2 0.437 on pIC50**.
- Built the **RDKit** preprocessing pipeline (SMILES standardisation, **2048-bit Morgan fingerprints**, **Tanimoto similarity**) for compound comparison and screening.
- Shipped a **Streamlit** app with multi-molecule prediction, ranking, explainability and applicability warnings, plus a **RAG literature assistant** (Europe PMC, TF-IDF, Gemini API).

Industrial Equipment Monitoring & IoT Platform **[GitHub]** MQTT, FastAPI, PostgreSQL, React, Docker

- Built a containerised IIoT platform ingesting **MQTT telemetry** (temperature, vibration, motor current, load, RPM) into PostgreSQL for real-time monitoring of simulated equipment.
- Delivered REST APIs and **WebSocket streaming** for live fleet status and historical telemetry analysis, with **threshold-based fault detection**, alerts and event tracking.
- Automated backend API and WebSocket tests and orchestrated all services with Docker Compose.

Aurora Beauty SEA Analytics Platform **[GitHub] [Live Demo]** Python, ETL, Streamlit, Dashboards

- Built ETL pipelines processing **50K+ daily transactions** across Southeast Asian markets, with KPI dashboards for demand forecasting and trend analysis and a hybrid recommendation engine.

RESEARCH EXPERIENCE

Geo-fencing & Boundary Surveillance using CCTV and IoT Sensors Nov 2025 – May 2026

Under the guidance of Dr. Susham Biswas, RGIPT

- Designed a real-time perimeter monitoring framework integrating CCTV, geo-fencing and IoT sensors, with polygon-based intrusion detection and automated alerts.
- Estimated speed, height and distance via camera calibration and computer vision, and built face-recognition workflows for long-range identity verification.

TECHNICAL SKILLS

Analytics & BI: SQL, Python, Pandas, NumPy, Power BI, Excel, EDA, Data Validation, Dimensional Modelling, ETL, Matplotlib
Databases & Engineering: PostgreSQL, MySQL, FastAPI, REST APIs, MQTT, WebSockets, Docker, Git, Linux
ML & AI: Scikit-learn, RDKit, OpenCV, YOLOv8, RAG, Streamlit

EDUCATION

Rajiv Gandhi Institute of Petroleum Technology (RGIPT), India 2021 – May 2026

Integrated Dual Degree: B.Tech in Computer Science & M.Tech in Artificial Intelligence